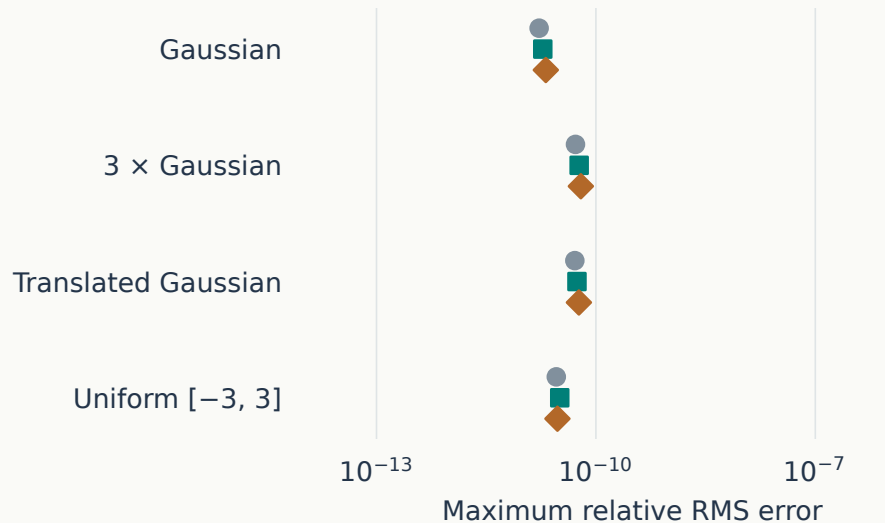


Derivative and circuit reconstruction errors

No gradient observations during learning · independent autograd and circuit checks

● Output ■ Gradient ◆ Hessian-vector

Derivative checks · 44 models



Qiskit round-trip · 11 teachers



Left: 512 inputs per distribution; each marker is the maximum across models. Right: eight translated inputs per circuit.

Dashed line: 10^{-6} . Circuit gradients use five-point finite differences. These are structured, classically simulable circuits.